Dennis Ulbrich

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Education

University of Bremen, Department of Mathematics, Mathematics — Bremen, Germany, Apr 2017 – Sept 2021

- Advisors: Prof. Dr. Jens Rademacher (University of Hamburg), Prof. Dr. Marc Kesseböhmer (University of Bremen)
- Referees: Prof. Dr. Jens Rademacher, Prof. Dr. Ian Melbourne (Warwick University)
- cf. MGP

University of Bremen, Department of Mathematics, Mathematics — Bremen, Germany, Apr 2016 – Apr 2017

- Further studies
- Extensions of the results of my Master thesis
- Successful acquisition of third-party funding (DFG) for PhD project

M.Sc. and B.Sc. **University of Bremen, Department of Mathematics**, Mathematics — Bremen, Germany, Oct 2011 – Apr 2016

- M.Sc. thesis: Dynamics of the three-state 1D Greenberg-Hastings cellular automaton
- Referees: Prof. Dr. Jens Rademacher, Dr. Tony Samuel (University of Exeter)
- B.Sc. thesis: Unerwartete Fehler bei bedingten Erwartungswerten und Wahrscheinlichkeiten
- Referees: Prof. Dr. Werner Brannath (University of Bremen), Dr. Kurt Falk (University of Kiel)

Experience

Hochschule Bremen, School of Electrical Engineering and Computer Science, Research assistant — Bremen, Germany, Feb 2025 – May 2025, 3 months

- Project work on AI-based transmission, analysis, verification of handwritten documents

Hochschule Bremen, School of Nature and Engineering, Lecturer in Mathematics — Bremen, Germany, Oct 2024 – Mar 2025, 6 months

- Mathematik 1

University of Münster, Department of Mathematics, Institute for Analysis and Numerics, Postdoctoral researcher — Münster, Germany, June 2023 – June 2024, 1 year 1 month

- Research on discrete hypocoercivity within DFG project 456849348
- Supervision: Prof. Dr. Marlies Pirner
- Keywords: Hypocoercivity, Kinetic equations, BGK-type approximations, Chemical reactions, Entropy methods

affinis AG, IT consultant — Bremen, Germany, Sept 2022 – Feb 2023, 6 months

Jacobs University Bremen (now: Constructor University Bremen), Lecturer in Mathematics — Bremen, Germany, Jan 2022 – June 2022, 6 months

- Finite Mathematics
- Introduction to Dynamical Systems

University of Bremen, Department of Mathematics, Research assistant (PhD student) — Bremen, Germany

Apr 2017 – Sept 2021
4 years 6 months

- Research within DFG project 384027439
- Research groups: Nonlinear Analysis and Applied Analysis, Stochastics and Dynamical Systems
- PhD thesis: Ergodic theory of nonlinear waves in discrete and continuous excitable media
- Advisors: Prof. Dr. Jens Rademacher (University of Hamburg), Prof. Dr. Marc Kesseböhmer (University of Bremen)
- Referees: Prof. Dr. Jens Rademacher, Prof. Dr. Ian Melbourne (Warwick University)
- Keywords: Nonlinear analysis, Ergodic Theory, PDE, Excitable media, Cellular automata, Dynamical Systems

Publications

Weak and strong interaction of excitation kinks in scalar parabolic equations
Published: 30 July 2021; Volume 35, pages 2199-2235, (2023) A. Pauthier, J. Rademacher, D. Ulbrich
A. Pauthier, J. Rademacher, D. Ulbrich
link.springer.com/article/10.1007/s10884-021-10040-2

Dynamics and topological entropy of 1D Greenberg-Hastings cellular automata
Published: May 2021, Volume 41, Issue 5, pages 1397 - 1430 M. Kesseböhmer, J. Rademacher, D. Ulbrich
M. Kesseböhmer, J. Rademacher, D. Ulbrich
www.cambridge.org/core/journals/ergodic-theory-and-dynamical-systems/article/dynamics-and-topological-entropy-of-1d-greenberghastings-cellular-automata/1DBE1EC4EC7BE9F216E9E35C214F2F3C

Ergodic theory of nonlinear waves in discrete and continuous excitable media
Diss., D. Ulbrich
D. Ulbrich
media.suub.uni-bremen.de/entities/publication/f6d1f31c-f9e0-4fcf-9d86-0ee66235e4d4

Discrete hypocoercivity for a nonlinear kinetic reaction model without initial close-to-equilibrium assumption (working title)
Ongoing work, together with L. Liu and M. Pirner
L. Liu, M. Pirner, D. Ulbrich

Grants

Brückenstipendium (funding period 12 months) — Apr 2016
The purpose of the bridging scholarship is to prepare for a doctoral project to be carried out at the University of Bremen and to secure the necessary external funding a scholarship/position - for example from one of the talent promotion organizations or through collaboration on a third-party funding application by the future supervisor.
University of Bremen
www.uni-bremen.de/fileadmin/user_upload/forschung/Anlage/Stipendium/Merkblatt_Brueckenstip.pdf

Academic Interests

Mathematics

Projects participation

DFG 456849348 — Jan 2021 – Dec 2026
Modeling and mathematical description of concrete physical applications in the context of kinetic theory using the Bathnagar-Gross-Krook equation
- Project details

DFG 384027439 Jan 2017 – Dec 2021

Ergodic theory of nonlinear waves in discrete and continuous excitable media

- Project details

IT-skills

LaTeX

Matlab, Mathematica, Java, C++, Oracle PL/SQL

Languages

German
Native speaker

English
Fluent

Other activities

Supervision

Organisation

Mathematical Collection